

Aenderungsbeschreibung/Description of Revision war kZ				Kommt vor/Frequency			
Buchst./Rev.Ltr. -.4		Erh-Antrags-Nr. / Promotion request No. PR085807		Aenderungs-Nr./Revision Notice No.		Bearbeitungsstatus/Lifecycle Freigegeben	
Changes from Var1 Ed 01.04 to Var1 Ed 01.06							
Corrections:							
Transfermode changed from 4 to 1 for 65172 HI/HHI P-Fuel Prefilter changed to LO/LOLO SPN 521595							
New measuring points added:							
PGN 65428 SPN 521057 Max Available Engine Power SPN 20 (in PGN 65172) Engine Coolant Pressure (Extended Range) PGN 64828 SPN 7101 Aftertreatment 1 Diesel Exhaust Fluid Pump SPN 166 (in PGN 65214) Engine Rated Power							
New limits added:							
-							
New alarms added:							
-							
Tolerierung/Tolerancing ISO 8015		Masse/Size ISO 14405					
Allgemeintoleranzen/General Tolerances ISO 2768-mK				Masse/Mass kg		Format/Size A4	
Bauart/Type GenolineNG		Allgemeine Fertigungsvorschriften nach MMN332 Production Specification per MMN 332		Werkstoff/Material			
Ausf. u.Lieferung Tech. Characteristics							
Oberflaechenschutz Surface Protection							
Verwendbar fuer Typ/Applicable to Model GLNG Release 30		Oberflaechenangaben nach MTN5033 Surface Specification per MTN5033		Halbzeug,Modell Gesenk / Semifinished Product Pattern, Die			
Projekt-/Auftrags-Nr./Project/Order No.		Fert.Ber. Man.Adv		Material-Nr./Material No. XZ00E52000037			
		Mont.Ber. Inst.Adv					
Ref.-Nr. / Ref. No. Type1 Var01 ED01_04		Emission ID				DOKUMENTATION GLNG Fremdschnittstelle J1939 DOCUMENTATION GLNG Foreign MCS Interface J1939	
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		Erstell. Drawn		30.03.2022		schaepers	
		Bearb. Change		31.03.2022		schaepers	
		Norm Std.		31.03.2022		ihmor stef	
		Gepr. Checked		31.03.2022		ihmor stef	
 Rolls-Royce Solutions GmbH		Zeichnungs-Nr./Drawing No. ZNG00045266				Blatt/ Sheet 1 von/of	
		Beschreibung/Description Type1 Var01 ED01_06				42	

Änderungsbeschreibung/Description of Revision war kZ			Kommt vor/Frequency
Buchst./Rev.Ltr. -4	Erh-Antrags-Nr. / Promotion request No. PR085807	Änderungs-Nr./Revision Notice No.	Bearbeitungsstatus/Lifecycle Freigegeben

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war kZ

Buchst./Rev.Ltr.
-4Erh-Antrags-Nr. / Promotion request No.
PR085807

Aenderungs-Nr./Revision Notice No.

Bearbeitungsstatus/Lifecycle
Freigegeben**J1939 PGN Tx Signals**

PGN-Name	PGN	PDUFormat	PDUspecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC
Engine Hours, Revolutions	65253	254	229	0	6	5	0	0	8
Exhaust Port Temperature 3	65185	254	161	0	7	1	1000	0	8
Dash Display	65276	254	252	0	6	1	1000	100	8
Engine Electronic Controller #1	61444	240	4	0	3	1	100	0	8
Engine Temperature 1	65262	254	238	0	6	1	1000	0	8
Fuel Information (Liquid)	65203	254	179	0	7	5	0	0	8
Exhaust Port Temperature 2	65186	254	162	0	7	1	1000	0	8
Exhaust Port Temperature 5	65183	254	159	0	7	1	1000	0	8
Turbocharger Information 1	65179	254	155	0	7	1	1000	0	8
MTU Air Start Pressure	65502	255	222	0	6	5	0	0	8
MTU Eng Contr 1	65360	255	80	0	3	4	100	100	8
Engine Configuration 1	65251	254	227	0	6	4	2000	100	39
Exhaust Temperature	65031	254	7	0	6	1	500	0	8
Trip Time Information 2	65200	254	176	0	7	5	0	0	20
Engine Fluid Level/Pressure #1	65263	254	239	0	6	1	500	0	8
MTU Eng Fluid Level Pressure 1	65363	255	83	0	6	1	500	0	8
MTU Eng Contr 4 - Addl Inform	65284	255	4	0	6	1	100	0	8
Fuel Consumption (Liquid)	65257	254	233	0	6	5	0	0	8
Engine Information	65170	254	146	0	7	1	100	0	8
Shutdown	65252	254	228	0	6	1	1000	0	8
MTU Eng Contr 2 - Speed Dem FB	65361	255	81	0	6	1	100	0	8
Electronic Engine Controller 2	61443	240	3	0	3	1	50	0	8
Exhaust Port Temperature 1	65187	254	163	0	7	1	1000	0	8
Fuel Economy (Liquid)	65266	254	242	0	6	1	100	0	8
Engine Temperature 2	65188	254	164	0	6	1	1000	0	8
Electronic Engine Controller 3	65247	254	223	0	6	1	250	0	8
Inlet/Exhaust Conditions 1	65270	254	246	0	6	1	500	0	8
Inlet/Exhaust Conditions 2	64976	253	208	0	6	1	500	0	8
Vehicle Electrical Power1	65271	254	247	0	6	1	1000	0	8
Electronic Engine Controller 4	65214	254	190	0	7	5	0	0	8
Engine Fuel / Lube Systems	65130	254	106	0	6	1	500	0	8
MTU Eng Contr 5 - Stat Signals	65367	255	87	0	6	1	500	0	8
MTU Engine Temperature 1	65364	255	84	0	6	1	1000	0	8
MTU Engine Temperature 2	65373	255	93	0	6	1	1000	0	8
Turbocharger	65245	254	221	0	6	1	1000	0	8
Exhaust Port Temperature 4	65184	254	160	0	7	1	1000	0	8
MTU Pressure Limits 1	65395	255	115	0	6	5	0	0	8
MTU Pressure Limits 2	65396	255	116	0	6	5	0	0	8
MTU Pressure Limits 3	65397	255	117	0	6	5	0	0	8
MTU Pressure Limits 4	65439	255	159	0	6	5	0	0	8
MTU Turbocharger 1 and 2 Speed Limits	65398	255	118	0	6	5	0	0	8
MTU Turbocharger 3 and 4 Speed Limits	65399	255	119	0	6	5	0	0	8
MTU Temperature Limits 1	65402	255	122	0	6	5	0	0	8
MTU Temperature Limits 2	65403	255	123	0	6	5	0	0	8
MTU-ETC-Information	65404	255	124	0	6	1	200	0	8
MTU Fuel Consumption Info	65405	255	125	0	6	1	500	0	8
MTU Splash Oil Temperature Limits 1	65406	255	126	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 2	65407	255	127	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 3	65408	255	128	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 4	65409	255	129	0	6	5	0	0	8
MTU Splash Oil Temperature Limits 5	65410	255	130	0	6	5	0	0	8
Main Bearing Temperature 1	65182	254	158	0	6	1	1000	0	8
Main Bearing Temperature 2	65181	254	157	0	6	1	1000	0	8
Main Bearing Temperature 3	65180	254	156	0	6	1	1000	0	8
Ambient Conditions	65269	254	245	0	6	1	1000	0	8
MTU Coolant Temperature (EMU) Limits	65413	255	133	0	6	5	0	0	8
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	255	132	0	6	5	0	0	8
Engine Auxiliary Coolant	65172	254	148	0	6	1	500	0	8
MTU Speed Limits	65415	255	135	0	6	5	0	0	8
MTU Refill Pump	65418	255	138	0	6	1	500	0	8
Time/Date	65254	254	230	0	6	5	0	0	8

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<table><thead><tr><th>PGN-Name</th><th>PGN</th><th>PDUFormat</th><th>PDUSpecific</th><th>DataPage</th><th>DefaultPriority</th><th>TransferMode</th><th>TxRepetitionRate</th><th>SendOnChangeTime</th><th>DLC</th></tr></thead><tbody><tr><td>MTU System DM1 Producer Extension</td><td>65483</td><td>255</td><td>203</td><td>0</td><td>6</td><td>4</td><td>1000</td><td>100</td><td>8</td></tr><tr><td>MTU User Information</td><td>65421</td><td>255</td><td>141</td><td>0</td><td>6</td><td>1</td><td>100</td><td>0</td><td>8</td></tr><tr><td>MTU Splash Oil Temperature 3</td><td>65371</td><td>255</td><td>91</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>Engine Average Information</td><td>64851</td><td>253</td><td>83</td><td>0</td><td>5</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>MTU Splash Oil Temperature 4</td><td>65370</td><td>255</td><td>90</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>MTU Exhaust Temperature Limits</td><td>65411</td><td>255</td><td>131</td><td>0</td><td>6</td><td>5</td><td>0</td><td>0</td><td>8</td></tr><tr><td>MTU Splash Oil Temperature 5</td><td>65369</td><td>255</td><td>89</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 Diesel Exhaust Fluid Tank Volume</td><td>65110</td><td>254</td><td>86</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>MTU Several Limits</td><td>65442</td><td>255</td><td>162</td><td>0</td><td>6</td><td>5</td><td>2000</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 Diesel Exhaust Fluid Information 1</td><td>64923</td><td>253</td><td>155</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 SCR Exhaust Gas Temperature 1</td><td>64830</td><td>253</td><td>62</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 2 SCR Exhaust Gas Temperature 1</td><td>64824</td><td>253</td><td>56</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>MTU SCR Limits</td><td>65443</td><td>255</td><td>163</td><td>0</td><td>6</td><td>5</td><td>2000</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 SCR Dosing System Information 1</td><td>61475</td><td>240</td><td>35</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 SCR Exhaust Gas Pressures</td><td>64831</td><td>253</td><td>63</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>EES / Electrical Load</td><td>65171</td><td>254</td><td>147</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>AT / Alternator Temperature</td><td>65191</td><td>254</td><td>167</td><td>0</td><td>7</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>Alarm Groups</td><td>65226</td><td>254</td><td>202</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>T-Exhaust Combined Limits</td><td>65400</td><td>255</td><td>120</td><td>0</td><td>6</td><td>5</td><td>0</td><td>0</td><td>8</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>255</td><td>139</td><td>0</td><td>6</td><td>5</td><td>0</td><td>0</td><td>24</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>255</td><td>140</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>MTU Generator Temperatures 1</td><td>65445</td><td>255</td><td>165</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>Dash Display 2</td><td>64671</td><td>252</td><td>159</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr><tr><td>MTU Several Limits 2</td><td>65448</td><td>255</td><td>168</td><td>0</td><td>6</td><td>5</td><td>2000</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 2 SCR Exhaust Gas Pressures</td><td>64825</td><td>253</td><td>57</td><td>0</td><td>6</td><td>1</td><td>500</td><td>0</td><td>8</td></tr><tr><td>MTU Energy Information</td><td>65428</td><td>255</td><td>148</td><td>0</td><td>6</td><td>1</td><td>50</td><td>0</td><td>8</td></tr><tr><td>Aftertreatment 1 Diesel Exhaust Fluid Supply Information</td><td>64828</td><td>253</td><td>60</td><td>0</td><td>6</td><td>1</td><td>1000</td><td>0</td><td>8</td></tr></tbody></table>											PGN-Name	PGN	PDUFormat	PDUSpecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC	MTU System DM1 Producer Extension	65483	255	203	0	6	4	1000	100	8	MTU User Information	65421	255	141	0	6	1	100	0	8	MTU Splash Oil Temperature 3	65371	255	91	0	6	1	1000	0	8	Engine Average Information	64851	253	83	0	5	1	500	0	8	MTU Splash Oil Temperature 4	65370	255	90	0	6	1	1000	0	8	MTU Exhaust Temperature Limits	65411	255	131	0	6	5	0	0	8	MTU Splash Oil Temperature 5	65369	255	89	0	6	1	1000	0	8	Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	65110	254	86	0	6	1	1000	0	8	MTU Several Limits	65442	255	162	0	6	5	2000	0	8	Aftertreatment 1 Diesel Exhaust Fluid Information 1	64923	253	155	0	6	1	500	0	8	Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	253	62	0	6	1	500	0	8	Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	253	56	0	6	1	500	0	8	MTU SCR Limits	65443	255	163	0	6	5	2000	0	8	Aftertreatment 1 SCR Dosing System Information 1	61475	240	35	0	6	1	500	0	8	Aftertreatment 1 SCR Exhaust Gas Pressures	64831	253	63	0	6	1	500	0	8	EES / Electrical Load	65171	254	147	0	6	1	500	0	8	AT / Alternator Temperature	65191	254	167	0	7	1	1000	0	8	Alarm Groups	65226	254	202	0	6	1	500	0	8	T-Exhaust Combined Limits	65400	255	120	0	6	5	0	0	8	T-Winding Limits	65419	255	139	0	6	5	0	0	24	MTU Binary Signals	65420	255	140	0	6	1	500	0	8	MTU Generator Temperatures 1	65445	255	165	0	6	1	1000	0	8	Dash Display 2	64671	252	159	0	6	1	1000	0	8	MTU Several Limits 2	65448	255	168	0	6	5	2000	0	8	Aftertreatment 2 SCR Exhaust Gas Pressures	64825	253	57	0	6	1	500	0	8	MTU Energy Information	65428	255	148	0	6	1	50	0	8	Aftertreatment 1 Diesel Exhaust Fluid Supply Information	64828	253	60	0	6	1	1000	0	8
PGN-Name	PGN	PDUFormat	PDUSpecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC																																																																																																																																																																																																																																																																																									
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MTU Splash Oil Temperature 3	65371	255	91	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
Engine Average Information	64851	253	83	0	5	1	500	0	8																																																																																																																																																																																																																																																																																									
MTU Splash Oil Temperature 4	65370	255	90	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
MTU Exhaust Temperature Limits	65411	255	131	0	6	5	0	0	8																																																																																																																																																																																																																																																																																									
MTU Splash Oil Temperature 5	65369	255	89	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	65110	254	86	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
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Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	253	56	0	6	1	500	0	8																																																																																																																																																																																																																																																																																									
MTU SCR Limits	65443	255	163	0	6	5	2000	0	8																																																																																																																																																																																																																																																																																									
Aftertreatment 1 SCR Dosing System Information 1	61475	240	35	0	6	1	500	0	8																																																																																																																																																																																																																																																																																									
Aftertreatment 1 SCR Exhaust Gas Pressures	64831	253	63	0	6	1	500	0	8																																																																																																																																																																																																																																																																																									
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Alarm Groups	65226	254	202	0	6	1	500	0	8																																																																																																																																																																																																																																																																																									
T-Exhaust Combined Limits	65400	255	120	0	6	5	0	0	8																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	255	139	0	6	5	0	0	24																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	255	140	0	6	1	500	0	8																																																																																																																																																																																																																																																																																									
MTU Generator Temperatures 1	65445	255	165	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
Dash Display 2	64671	252	159	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
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MTU Energy Information	65428	255	148	0	6	1	50	0	8																																																																																																																																																																																																																																																																																									
Aftertreatment 1 Diesel Exhaust Fluid Supply Information	64828	253	60	0	6	1	1000	0	8																																																																																																																																																																																																																																																																																									
 A Rolls-Royce solution Rolls-Royce Solutions GmbH		Zeichnungs-Nr./Drawing No.		ZNG00045266				Blatt/ Sheet 4 von/of 42																																																																																																																																																																																																																																																																																										
		Material-Nr./Material No.		XZ00E52000037																																																																																																																																																																																																																																																																																														
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J1939 PGN Rx Signals

PGN-Name	PGN	PDUFormat	PDUSpecific	DataPage	DefaultPriority	TransferMode	TxRepetitionRate	SendOnChangeTime	DLC
MTU System Operating PDU1 prop A1	61184	239	0	0	6	4	2000	100	8

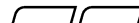
J1939 SPN Tx Signals

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
Engine Hours, Revolutions	65253	Engine Total Hours of Operation	247	32	1	1	58
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 9 Temperature	1145	16	1	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 10 Temperature	1146	16	3	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 11 Temperature	1147	16	5	1	68
Exhaust Port Temperature 3	65185	Engine Exhaust Gas Port 12 Temperature	1148	16	7	1	68
Dash Display	65276	Engine Fuel Filter Differential Pressure	95	8	3	1	53
Dash Display	65276	Engine Oil Filter Differential Pressure	99	8	4	1	54
Engine Electronic Controller #1	61444	Engine Speed	190	16	4	1	76
Engine Temperature 1	65262	Engine Coolant Temperature	110	8	1	1	67
Engine Temperature 1	65262	Engine Fuel 1 Temperature 1	174	8	2	1	67
Engine Temperature 1	65262	Engine Oil Temperature 1	175	16	3	1	68
Engine Temperature 1	65262	Engine Turbocharger Oil Temperature	176	16	5	1	68
Fuel Information (Liquid)	65203	Trip Average Fuel Rate	1029	16	5	1	42
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 5 Temperature	1141	16	1	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 6 Temperature	1142	16	3	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 7 Temperature	1143	16	5	1	68
Exhaust Port Temperature 2	65186	Engine Exhaust Gas Port 8 Temperature	1144	16	7	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 17 Temperature	1153	16	1	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 18 Temperature	1154	16	3	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 19 Temperature	1155	16	5	1	68
Exhaust Port Temperature 5	65183	Engine Exhaust Gas Port 20 Temperature	1156	16	7	1	68
Turbocharger Information 1	65179	Engine Turbocharger Lube Oil Pressure 2	1168	8	1	1	47
Turbocharger Information 1	65179	Engine Turbocharger 2 Speed	1169	16	2	1	77
Turbocharger Information 1	65179	Engine Turbocharger 3 Speed	1170	16	4	1	77
Turbocharger Information 1	65179	Engine Turbocharger 4 Speed	1171	16	6	1	77
MTU Air Start Pressure	65502	MTU Engine Air Start Pressure (Extended Range)	520274	16	1	1	106
MTU Eng Contr 1	65360	MTU Engine Speed redundant	520382	16	1	1	76
MTU Eng Contr 1	65360	Engine Turning Status	520200	2	3	1	87
MTU Eng Contr 1	65360	Engine Safety and Protection Override Status	520202	2	3	3	87
MTU Eng Contr 1	65360	Engine Local Operation Status	520203	2	3	5	87
MTU Eng Contr 1	65360	MTU Engine Running State	520255	2	3	7	87
MTU Eng Contr 1	65360	Engine Cylinder Cutoff	520252	2	5	1	87
MTU Eng Contr 1	65360	Engine Turbocharger 2 Status	520237	2	5	3	87
MTU Eng Contr 1	65360	Engine Turbocharger 3 Status	520242	2	8	1	87
MTU Eng Contr 1	65360	Engine Turbocharger 4 Status	520243	2	8	3	87
Engine Configuration 1	65251	Engine Speed At Idle, Point 1	188	16	1	1	76
Exhaust Temperature	65031	Engine Exhaust Manifold Bank 2 Temperature 1	2433	16	1	1	68
Exhaust Temperature	65031	Engine Exhaust Manifold Bank 1 Temperature 1	2434	16	3	1	68
Trip Time Information 2	65200	Trip Engine Running Time	1036	32	9	1	58
Engine Fluid Level/Pressure #1	65263	Engine Fuel Delivery Pressure	94	8	1	1	47
Engine Fluid Level/Pressure #1	65263	Engine Oil Pressure	100	8	4	1	47
Engine Fluid Level/Pressure #1	65263	Engine Crankcase Pressure 1	101	16	5	1	52
Engine Fluid Level/Pressure #1	65263	Engine Coolant Pressure 1	109	8	7	1	53
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Oil Pressure redundant	520292	8	1	1	47
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Fuel Delivery Pressure behind Filter	520286	8	2	1	47
MTU Eng Fluid Level Pressure 1	65363	MTU Engine Fuel Pre-Filter Differential Pressure	520287	8	3	1	53
MTU Eng Contr 4 - Addl Inform	65284	MTU Engine Controller Error Codes	520256	16	1	1	12
Fuel Consumption (Liquid)	65257	Engine Trip Fuel	182	32	1	1	20
Fuel Consumption (Liquid)	65257	Engine Total Fuel Used	250	32	5	1	20
Engine Information	65170	Engine Oil Filter Intake Pressure	1208	8	1	1	47
Shutdown	65252	Engine Protection System has Shutdown Engine	1110	2	5	1	87

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	Material-Nr./Material No.	XZ00E52000037	
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Aenderungsbeschreibung/Description of Revision war kZ			Kommt vor/Frequency
Buchst./Rev.Ltr. -4	Erh-Antrags-Nr. / Promotion request No. PR085807	Aenderungs-Nr./Revision Notice No.	Bearbeitungsstatus/Lifecycle Freigegeben

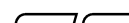
PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
Shutdown	65252	Engine Protection System Approaching Shutdown	1109	2	5	3	87
Shutdown	65252	Engine Overspeed Test	2812	2	7	7	87
MTU Eng Contr 2 - Speed Dem FB	65361	Engine Demanded Operating Speed	520707	16	1	1	76
Electronic Engine Controller 2	61443	Engine Percent Load At Current Speed	92	8	3	1	46
Electronic Engine Controller 2	61443	Actual Maximum Available Engine - Percent Torque	3357	8	7	1	44
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 1 Temperature	1137	16	1	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 2 Temperature	1138	16	3	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 3 Temperature	1139	16	5	1	68
Exhaust Port Temperature 1	65187	Engine Exhaust Gas Port 4 Temperature	1140	16	7	1	68
Fuel Economy (Liquid)	65266	Engine Fuel Rate	183	16	1	1	42
Engine Temperature 2	65188	Engine ECU Temperature	1136	16	3	1	68
Electronic Engine Controller 3	65247	Engine's Desired Operating Speed	515	16	2	1	76
Inlet/Exhaust Conditions 1	65270	Engine Intake Manifold 1 Temperature	105	8	3	1	67
Inlet/Exhaust Conditions 2	64976	Engine Intake Manifold #1 Absolute Pressure	3563	8	5	1	53
Vehicle Electrical Power1	65271	Battery Potential / Power Input 1	168	16	5	1	80
Vehicle Electrical Power1	65271	Key Switch Battery Potential	158	16	7	1	80
Electronic Engine Controller 4	65214	Engine Rated Power	166	16	1	1	55
Electronic Engine Controller 4	65214	Engine Rated Speed	189	16	3	1	76
Engine Fuel / Lube Systems	65130	Engine Oil Priming Pump Switch	3550	2	6	1	87
MTU Eng Contr 5 - Stat Signals	65367	Engine Status of Startprocedure	520241	32	3	1	245
MTU Eng Contr 5 - Stat Signals	65367	MTU Start Interlock	520266	2	7	1	87
MTU Engine Temperature 1	65364	MTU Engine Coolant Temperature redundant	520302	8	1	1	67
MTU Engine Temperature 1	65364	Engine Charge Air Sequential Control Valve Temperature	520254	16	2	1	68
MTU Engine Temperature 1	65364	Engine Main Bearing Temperature Average	520407	16	4	1	68
MTU Engine Temperature 2	65373	Engine Exhaust Gas Temperature - Left 2	520393	16	1	1	68
MTU Engine Temperature 2	65373	Engine Exhaust Gas Temperature - Right 2	520394	16	3	1	68
Turbocharger	65245	Engine Turbocharger Lube Oil Pressure 1	104	8	1	1	47
Turbocharger	65245	Engine Turbocharger 1 Speed	103	16	2	1	77
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 13 Temperature	1149	16	1	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 14 Temperature	1150	16	3	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 15 Temperature	1151	16	5	1	68
Exhaust Port Temperature 4	65184	Engine Exhaust Gas Port 16 Temperature	1152	16	7	1	68
MTU Pressure Limits 1	65395	LM LO P-Lube Oil	520500	8	1	1	47
MTU Pressure Limits 1	65395	LM LOLO P-Lube Oil	520501	8	2	1	47
MTU Pressure Limits 1	65395	LM HI P-Crankcase	520502	16	3	1	52
MTU Pressure Limits 1	65395	LM HIHI P-Crankcase	520503	16	5	1	52
MTU Pressure Limits 1	65395	LM LO P-Coolant	520504	8	7	1	53
MTU Pressure Limits 1	65395	LM LOLO P-Coolant	520505	8	8	1	53
MTU Pressure Limits 2	65396	LM LO P-Fuel	520506	8	1	1	47
MTU Pressure Limits 2	65396	LM LOLO P-Fuel	520507	8	2	1	47
MTU Pressure Limits 2	65396	LM HI P-Charge Air	520508	8	3	1	53
MTU Pressure Limits 2	65396	LM HI P-Lube Oil Filter Diff.	520537	8	5	1	54
MTU Pressure Limits 2	65396	LM HI P-Fuel Filter Diff.	520538	8	6	1	54
MTU Pressure Limits 2	65396	LM LO Limit Engine Turbocharger Lube Oil Pressure 1	520470	8	8	1	47
MTU Pressure Limits 3	65397	LM LO P-Start Air	520540	16	1	1	53
MTU Pressure Limits 3	65397	LM LOLO P-Start Air Start Term	520541	16	3	1	47
MTU Pressure Limits 3	65397	LM LO P-Oil Refill Pump	520542	8	5	1	47
MTU Pressure Limits 3	65397	LM LOLO Limit Engine Turbocharger Lube Oil Pressure 1	520471	8	6	1	47
MTU Pressure Limits 3	65397	LM LO Limit Engine Turbocharger Lube Oil Pressure 2	520472	8	7	1	47
MTU Pressure Limits 3	65397	LM LOLO Limit Engine Turbocharger Lube Oil Pressure 2	520473	8	8	1	47
MTU Pressure Limits 4	65439	LM HIHI Lube Oil Filter Diff Pressure	521580	8	3	1	47
MTU Pressure Limits 4	65439	LM LOLO P-Raw Water	521932	8	4	1	53
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HI ETC 1 Speed	520510	16	1	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HIHI ETC 1 Speed	520511	16	3	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HI ETC 2 Speed	520512	16	5	1	77
MTU Turbocharger 1 and 2 Speed Limits	65398	LM HIHI ETC 2 Speed	520513	16	7	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HI ETC 3 Speed	520514	16	1	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HIHI ETC 3 Speed	520515	16	3	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HI ETC 4 Speed	520516	16	5	1	77
MTU Turbocharger 3 and 4 Speed Limits	65399	LM HIHI ETC 4 Speed	520517	16	7	1	77

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	Material-Nr./Material No.	XZ00E52000037	
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Aenderungsbeschreibung/Description of Revision war kZ					Kommt vor/Frequency																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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<table><tr><th>PGN-Name</th><th>PGN</th><th>SPN-Name</th><th>SPN</th><th>BitSize</th><th>StartByte</th><th>StartBit</th><th>SLOT</th></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HI T-Coolant (ECU)</td><td>520518</td><td>8</td><td>1</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HIHI T-Coolant (ECU)</td><td>520519</td><td>8</td><td>2</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HI T-Charge Air</td><td>520520</td><td>8</td><td>3</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HIHI T-Charge Air</td><td>520521</td><td>8</td><td>4</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HI T-Lube Oil</td><td>520522</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Temperature Limits 1</td><td>65402</td><td>LM HIHI T-Lube Oil</td><td>520523</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU Temperature Limits 2</td><td>65403</td><td>LM HI T-Charge Air Seq Ctrl Valve</td><td>520532</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Temperature Limits 2</td><td>65403</td><td>LM HI T-Fuel</td><td>520534</td><td>8</td><td>3</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 2</td><td>65403</td><td>LM HIHI T-Fuel</td><td>520535</td><td>8</td><td>4</td><td>1</td><td>67</td></tr><tr><td>MTU Temperature Limits 2</td><td>65403</td><td>UL T-ECU</td><td>520536</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Temperature Limits 2</td><td>65403</td><td>LM HIHI T-Charge Air Seq Ctrl Valve</td><td>520533</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU-ETC-Information</td><td>65404</td><td>ETC 1 Running</td><td>520560</td><td>2</td><td>2</td><td>1</td><td>87</td></tr><tr><td>MTU-ETC-Information</td><td>65404</td><td>ETC 2 Running</td><td>520561</td><td>2</td><td>2</td><td>3</td><td>87</td></tr><tr><td>MTU-ETC-Information</td><td>65404</td><td>ETC 3 Running</td><td>520562</td><td>2</td><td>2</td><td>5</td><td>87</td></tr><tr><td>MTU-ETC-Information</td><td>65404</td><td>ETC 4 Running</td><td>520563</td><td>2</td><td>2</td><td>7</td><td>87</td></tr><tr><td>MTU Fuel Consumption Info</td><td>65405</td><td>Max. Scale Fuel Consumpt./h</td><td>520564</td><td>16</td><td>1</td><td>1</td><td>42</td></tr><tr><td>MTU Fuel Consumption Info</td><td>65405</td><td>Actual Fuel Cons. %/l</td><td>520565</td><td>8</td><td>3</td><td>1</td><td>44</td></tr><tr><td>MTU Fuel Consumption Info</td><td>65405</td><td>Mean Trip Fuel Cons. %/l</td><td>520566</td><td>8</td><td>4</td><td>1</td><td>44</td></tr><tr><td>MTU Splash Oil Temperature Limits 1</td><td>65406</td><td>LM HI T-Splash Oil B1</td><td>520362</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 1</td><td>65406</td><td>LM HIHI T-Splash Oil B1</td><td>520363</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 1</td><td>65406</td><td>LM HI T-Splash Oil B2</td><td>520364</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 1</td><td>65406</td><td>LM HIHI T-Splash Oil B2</td><td>520365</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 2</td><td>65407</td><td>LM HI T-Splash Oil B3</td><td>520366</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 2</td><td>65407</td><td>LM HIHI T-Splash Oil B3</td><td>520367</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 2</td><td>65407</td><td>LM HI T-Splash Oil B4</td><td>520368</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 2</td><td>65407</td><td>LM HIHI T-Splash Oil B4</td><td>520369</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 3</td><td>65408</td><td>LM HI T-Splash Oil B5</td><td>520370</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 3</td><td>65408</td><td>LM HIHI T-Splash Oil B5</td><td>520371</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 3</td><td>65408</td><td>LM HI T-Splash Oil B6</td><td>520372</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 3</td><td>65408</td><td>LM HIHI T-Splash Oil B6</td><td>520373</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 4</td><td>65409</td><td>LM HI T-Splash Oil B7</td><td>520374</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 4</td><td>65409</td><td>LM HIHI T-Splash Oil B7</td><td>520375</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 4</td><td>65409</td><td>LM HI T-Splash Oil B8</td><td>520376</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 4</td><td>65409</td><td>LM HIHI T-Splash Oil B8</td><td>520377</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 5</td><td>65410</td><td>LM HI T-Splash Oil B9</td><td>520378</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 5</td><td>65410</td><td>LM HIHI T-Splash Oil B9</td><td>520379</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 5</td><td>65410</td><td>LM HI T-Splash Oil B10</td><td>520380</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Splash Oil Temperature Limits 5</td><td>65410</td><td>LM HIHI T-Splash Oil B10</td><td>520381</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 1</td><td>65182</td><td>Engine Main Bearing 1 Temperature</td><td>1157</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 1</td><td>65182</td><td>Engine Main Bearing 2 Temperature</td><td>1158</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 1</td><td>65182</td><td>Engine Main Bearing 3 Temperature</td><td>1159</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 1</td><td>65182</td><td>Engine Main Bearing 4 Temperature</td><td>1160</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 2</td><td>65181</td><td>Engine Main Bearing 5 Temperature</td><td>1161</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 2</td><td>65181</td><td>Engine Main Bearing 6 Temperature</td><td>1162</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 2</td><td>65181</td><td>Engine Main Bearing 7 Temperature</td><td>1163</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 2</td><td>65181</td><td>Engine Main Bearing 8 Temperature</td><td>1164</td><td>16</td><td>7</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 3</td><td>65180</td><td>Engine Main Bearing 9 Temperature</td><td>1165</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 3</td><td>65180</td><td>Engine Main Bearing 10 Temperature</td><td>1166</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>Main Bearing Temperature 3</td><td>65180</td><td>Engine Main Bearing 11 Temperature</td><td>1167</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>Ambient Conditions</td><td>65269</td><td>Engine Intake 1 Air Temperature</td><td>172</td><td>8</td><td>6</td><td>1</td><td>67</td></tr><tr><td>MTU Coolant Temperature (EMU) Limits</td><td>65413</td><td>LM HI T-Coolant (EMU)</td><td>520352</td><td>8</td><td>1</td><td>1</td><td>67</td></tr><tr><td>MTU Coolant Temperature (EMU) Limits</td><td>65413</td><td>LM HIHI T-Coolant (EMU)</td><td>520353</td><td>8</td><td>2</td><td>1</td><td>67</td></tr><tr><td>MTU Coolant Temperature (EMU) Limits</td><td>65413</td><td>LM HI Engine Air Inlet Temperature</td><td>520475</td><td>8</td><td>3</td><td>1</td><td>67</td></tr><tr><td>MTU Bearing Temperature and Raw-Water Pressure Limits</td><td>65412</td><td>LM HI T-Bearing Mean</td><td>520358</td><td>16</td><td>1</td><td>1</td><td>68</td></tr><tr><td>MTU Bearing Temperature and Raw-Water Pressure Limits</td><td>65412</td><td>LM HI T-Bearing</td><td>520359</td><td>16</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Bearing Temperature and Raw-Water Pressure Limits</td><td>65412</td><td>LM HIHI T-Bearing</td><td>520360</td><td>16</td><td>5</td><td>1</td><td>68</td></tr><tr><td>MTU Bearing Temperature and Raw-Water Pressure Limits</td><td>65412</td><td>LM LO P-Raw Water</td><td>520361</td><td>8</td><td>7</td><td>1</td><td>53</td></tr><tr><td>MTU Bearing Temperature and Raw-Water Pressure Limits</td><td></td><td>LM HIHI P-Crankcase</td><td>520474</td><td>8</td><td>8</td><td>1</td><td>53</td></tr><tr><td>Engine Auxiliary Coolant</td><td>65172</td><td>Sea Water Pump Outlet Pressure</td><td>2435</td><td>8</td><td>3</td><td>1</td><td>53</td></tr><tr><td>Engine Auxiliary Coolant</td><td>65172</td><td>Engine Coolant Pressure 1 (Extended Range)</td><td>20</td><td>8</td><td>4</td><td>1</td><td>47</td></tr><tr><td>MTU Speed Limits</td><td>65415</td><td>LM HIHI Engine Speed</td><td>520387</td><td>16</td><td>1</td><td>1</td><td>76</td></tr><tr><td>MTU Speed Limits</td><td>65415</td><td>UL Engine Speed (EMU)</td><td>520725</td><td>16</td><td>3</td><td>1</td><td>76</td></tr><tr><td>MTU Refill Pump</td><td>65418</td><td>P-Oil Refill Pump</td><td>520567</td><td>8</td><td>1</td><td>1</td><td>47</td></tr></table>									PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT	MTU Temperature Limits 1	65402	LM HI T-Coolant (ECU)	520518	8	1	1	67	MTU Temperature Limits 1	65402	LM HIHI T-Coolant (ECU)	520519	8	2	1	67	MTU Temperature Limits 1	65402	LM HI T-Charge Air	520520	8	3	1	67	MTU Temperature Limits 1	65402	LM HIHI T-Charge Air	520521	8	4	1	67	MTU Temperature Limits 1	65402	LM HI T-Lube Oil	520522	16	5	1	68	MTU Temperature Limits 1	65402	LM HIHI T-Lube Oil	520523	16	7	1	68	MTU Temperature Limits 2	65403	LM HI T-Charge Air Seq Ctrl Valve	520532	16	1	1	68	MTU Temperature Limits 2	65403	LM HI T-Fuel	520534	8	3	1	67	MTU Temperature Limits 2	65403	LM HIHI T-Fuel	520535	8	4	1	67	MTU Temperature Limits 2	65403	UL T-ECU	520536	16	5	1	68	MTU Temperature Limits 2	65403	LM HIHI T-Charge Air Seq Ctrl Valve	520533	16	7	1	68	MTU-ETC-Information	65404	ETC 1 Running	520560	2	2	1	87	MTU-ETC-Information	65404	ETC 2 Running	520561	2	2	3	87	MTU-ETC-Information	65404	ETC 3 Running	520562	2	2	5	87	MTU-ETC-Information	65404	ETC 4 Running	520563	2	2	7	87	MTU Fuel Consumption Info	65405	Max. Scale Fuel Consumpt./h	520564	16	1	1	42	MTU Fuel Consumption Info	65405	Actual Fuel Cons. %/l	520565	8	3	1	44	MTU Fuel Consumption Info	65405	Mean Trip Fuel Cons. %/l	520566	8	4	1	44	MTU Splash Oil Temperature Limits 1	65406	LM HI T-Splash Oil B1	520362	16	1	1	68	MTU Splash Oil Temperature Limits 1	65406	LM HIHI T-Splash Oil B1	520363	16	3	1	68	MTU Splash Oil Temperature Limits 1	65406	LM HI T-Splash Oil B2	520364	16	5	1	68	MTU Splash Oil Temperature Limits 1	65406	LM HIHI T-Splash Oil B2	520365	16	7	1	68	MTU Splash Oil Temperature Limits 2	65407	LM HI T-Splash Oil B3	520366	16	1	1	68	MTU Splash Oil Temperature Limits 2	65407	LM HIHI T-Splash Oil B3	520367	16	3	1	68	MTU Splash Oil Temperature Limits 2	65407	LM HI T-Splash Oil B4	520368	16	5	1	68	MTU Splash Oil Temperature Limits 2	65407	LM HIHI T-Splash Oil B4	520369	16	7	1	68	MTU Splash Oil Temperature Limits 3	65408	LM HI T-Splash Oil B5	520370	16	1	1	68	MTU Splash Oil Temperature Limits 3	65408	LM HIHI T-Splash Oil B5	520371	16	3	1	68	MTU Splash Oil Temperature Limits 3	65408	LM HI T-Splash Oil B6	520372	16	5	1	68	MTU Splash Oil Temperature Limits 3	65408	LM HIHI T-Splash Oil B6	520373	16	7	1	68	MTU Splash Oil Temperature Limits 4	65409	LM HI T-Splash Oil B7	520374	16	1	1	68	MTU Splash Oil Temperature Limits 4	65409	LM HIHI T-Splash Oil B7	520375	16	3	1	68	MTU Splash Oil Temperature Limits 4	65409	LM HI T-Splash Oil B8	520376	16	5	1	68	MTU Splash Oil Temperature Limits 4	65409	LM HIHI T-Splash Oil B8	520377	16	7	1	68	MTU Splash Oil Temperature Limits 5	65410	LM HI T-Splash Oil B9	520378	16	1	1	68	MTU Splash Oil Temperature Limits 5	65410	LM HIHI T-Splash Oil B9	520379	16	3	1	68	MTU Splash Oil Temperature Limits 5	65410	LM HI T-Splash Oil B10	520380	16	5	1	68	MTU Splash Oil Temperature Limits 5	65410	LM HIHI T-Splash Oil B10	520381	16	7	1	68	Main Bearing Temperature 1	65182	Engine Main Bearing 1 Temperature	1157	16	1	1	68	Main Bearing Temperature 1	65182	Engine Main Bearing 2 Temperature	1158	16	3	1	68	Main Bearing Temperature 1	65182	Engine Main Bearing 3 Temperature	1159	16	5	1	68	Main Bearing Temperature 1	65182	Engine Main Bearing 4 Temperature	1160	16	7	1	68	Main Bearing Temperature 2	65181	Engine Main Bearing 5 Temperature	1161	16	1	1	68	Main Bearing Temperature 2	65181	Engine Main Bearing 6 Temperature	1162	16	3	1	68	Main Bearing Temperature 2	65181	Engine Main Bearing 7 Temperature	1163	16	5	1	68	Main Bearing Temperature 2	65181	Engine Main Bearing 8 Temperature	1164	16	7	1	68	Main Bearing Temperature 3	65180	Engine Main Bearing 9 Temperature	1165	16	1	1	68	Main Bearing Temperature 3	65180	Engine Main Bearing 10 Temperature	1166	16	3	1	68	Main Bearing Temperature 3	65180	Engine Main Bearing 11 Temperature	1167	16	5	1	68	Ambient Conditions	65269	Engine Intake 1 Air Temperature	172	8	6	1	67	MTU Coolant Temperature (EMU) Limits	65413	LM HI T-Coolant (EMU)	520352	8	1	1	67	MTU Coolant Temperature (EMU) Limits	65413	LM HIHI T-Coolant (EMU)	520353	8	2	1	67	MTU Coolant Temperature (EMU) Limits	65413	LM HI Engine Air Inlet Temperature	520475	8	3	1	67	MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing Mean	520358	16	1	1	68	MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing	520359	16	3	1	68	MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HIHI T-Bearing	520360	16	5	1	68	MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM LO P-Raw Water	520361	8	7	1	53	MTU Bearing Temperature and Raw-Water Pressure Limits		LM HIHI P-Crankcase	520474	8	8	1	53	Engine Auxiliary Coolant	65172	Sea Water Pump Outlet Pressure	2435	8	3	1	53	Engine Auxiliary Coolant	65172	Engine Coolant Pressure 1 (Extended Range)	20	8	4	1	47	MTU Speed Limits	65415	LM HIHI Engine Speed	520387	16	1	1	76	MTU Speed Limits	65415	UL Engine Speed (EMU)	520725	16	3	1	76	MTU Refill Pump	65418	P-Oil Refill Pump	520567	8	1	1	47
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MTU Fuel Consumption Info	65405	Actual Fuel Cons. %/l	520565	8	3	1	44																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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Main Bearing Temperature 3	65180	Engine Main Bearing 11 Temperature	1167	16	5	1	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
Ambient Conditions	65269	Engine Intake 1 Air Temperature	172	8	6	1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Coolant Temperature (EMU) Limits	65413	LM HI T-Coolant (EMU)	520352	8	1	1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Coolant Temperature (EMU) Limits	65413	LM HIHI T-Coolant (EMU)	520353	8	2	1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Coolant Temperature (EMU) Limits	65413	LM HI Engine Air Inlet Temperature	520475	8	3	1	67																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing Mean	520358	16	1	1	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HI T-Bearing	520359	16	3	1	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM HIHI T-Bearing	520360	16	5	1	68																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Bearing Temperature and Raw-Water Pressure Limits	65412	LM LO P-Raw Water	520361	8	7	1	53																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Bearing Temperature and Raw-Water Pressure Limits		LM HIHI P-Crankcase	520474	8	8	1	53																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
Engine Auxiliary Coolant	65172	Sea Water Pump Outlet Pressure	2435	8	3	1	53																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
Engine Auxiliary Coolant	65172	Engine Coolant Pressure 1 (Extended Range)	20	8	4	1	47																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Speed Limits	65415	LM HIHI Engine Speed	520387	16	1	1	76																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Speed Limits	65415	UL Engine Speed (EMU)	520725	16	3	1	76																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
MTU Refill Pump	65418	P-Oil Refill Pump	520567	8	1	1	47																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
 A Rolls-Royce solution Rolls-Royce Solutions GmbH		Zeichnungs-Nr./Drawing No. Material-Nr./Material No.		ZNG00045266 XZ00E52000037				Blatt/ Sheet 7 von/of 42																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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Aenderungsbeschreibung/Description of Revision war kZ			Kommt vor/Frequency
Buchst./Rev.Ltr. -4	Erh-Antrags-Nr. / Promotion request No. PR085807	Aenderungs-Nr./Revision Notice No.	Bearbeitungsstatus/Lifecycle Freigegeben

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
MTU Refill Pump	65418	P-Fuel Prefilter	521595	8	3	1	MTU_pr01
MTU Refill Pump	65418	P-RA Filter Out at PFM	521438	16	4	1	85
Time/Date	65254	Seconds	959	8	1	1	63
Time/Date	65254	Minutes	960	8	2	1	61
Time/Date	65254	Hours	961	8	3	1	3
Time/Date	65254	Month	963	8	4	1	62
Time/Date	65254	Day	962	8	5	1	60
Time/Date	65254	Year	964	8	6	1	66
MTU System DM1 Producer Extension	65483	MTU DM1 Horn Status	520815	2	1	1	87
MTU User Information	65421	User Information	520714	16	1	1	276
MTU Splash Oil Temperature 3	65371	MTU Splash Oil Temperature Mean	520467	16	1	1	68
MTU Splash Oil Temperature 3	65371	Splash Oil Temperature 11	520395	16	5	1	68
MTU Splash Oil Temperature 3	65371	Splash Oil Temperature 12	520396	16	7	1	68
Engine Average Information	64851	Engine Exhaust Temperature Average	4151	16	1	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 13	520397	16	1	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 14	520398	16	3	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 15	520399	16	5	1	68
MTU Splash Oil Temperature 4	65370	Splash Oil Temperature 16	520400	16	7	1	68
MTU Exhaust Temperature Limits	65411	LM HI T-Exhaust	520354	16	1	1	68
MTU Exhaust Temperature Limits	65411	LM LO T-Exhaust	520355	16	3	1	68
MTU Exhaust Temperature Limits	65411	LM HIHI T-Exhaust	520356	16	5	1	68
MTU Exhaust Temperature Limits	65411	LM LOLO T-Exhaust	520357	16	7	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 17	520401	16	1	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 18	520402	16	3	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 19	520403	16	5	1	68
MTU Splash Oil Temperature 5	65369	Splash Oil Temperature 20	520404	16	7	1	68
Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	65110	Aftertreatment 1 Diesel Exhaust Fluid Tank Volume	1761	8	1	1	44
MTU Several Limits	65442	LM LO P-RA Filter Out at PFM	521586	16	1	1	85
MTU Several Limits	65442	LM HI T-RA at PFM	521587	8	3	1	67
MTU Several Limits	65442	LM LO T-RA at PFM	521588	8	4	1	67
MTU Several Limits	65442	LM LO RA Tank Level	521589	8	5	1	44
MTU Several Limits	65442	LM LOLO RA Tank Level	521590	8	6	1	44
MTU Several Limits	65442	LM LO P-Fuel Prefilter	521591	8	7	1	MTU_pr01
MTU Several Limits	65442	LM LOLO P-Fuel Prefilter	521592	8	8	1	MTU_pr01
Aftertreatment 1 Diesel Exhaust Fluid Information 1	64923	Aftertreatment 1 Diesel Exhaust Fluid Temperature 2	3515	8	1	1	67
Aftertreatment 1 Diesel Exhaust Fluid Information 1	64923	Aftertreatment 1 Diesel Exhaust Fluid Concentration	3516	8	2	1	254
Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	Aftertreatment 1 SCR Intake Temperature	4360	16	1	1	68
Aftertreatment 1 SCR Exhaust Gas Temperature 1	64830	Aftertreatment 1 SCR Outlet Temperature	4363	16	4	1	68
Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	Aftertreatment 2 SCR Intake Temperature	4413	16	1	1	68
Aftertreatment 2 SCR Exhaust Gas Temperature 1	64824	Aftertreatment 2 SCR Outlet Temperature	4415	16	4	1	68
MTU SCR Limits	65443	LM HI T-Exhaust SCR	521582	16	1	1	68
MTU SCR Limits	65443	LM HIHI T-Exhaust SCR	521583	16	3	1	68
MTU SCR Limits	65443	LM HI P-Diff Exhaust SCR 1	521584	16	5	1	85
MTU SCR Limits	65443	LM HIHI P-Diff Exhaust SCR 1	521585	16	7	1	85
Aftertreatment 1 SCR Dosing System Information 1	61475	Aftertreatment 1 SCR System 1 State	4332	4	3	1	89
Aftertreatment 1 SCR Exhaust Gas Pressures	64831	Aftertreatment 1 SCR Differential Pressure	4358	16	1	1	85
Aftertreatment 1 SCR Exhaust Gas Pressures	64831	Aftertreatment 1 SCR Intake Pressure	6586	16	4	1	355
EES / Electrical Load	65171	Electrical Load	1204	16	1	1	55
AT / Alternator Temperature	65191	Engine Alternator Bearing 1 Temperature	1122	8	1	1	67
AT / Alternator Temperature	65191	Engine Alternator Bearing 2 Temperature	1123	8	2	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 1 Temperature	1124	8	3	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 2 Temperature	1125	8	4	1	67
AT / Alternator Temperature	65191	Engine Alternator Winding 3 Temperature	1126	8	5	1	67
T-Exhaust Combined Limits	65400	LM HI T-Exhaust Combined A	520524	16	1	1	68
T-Exhaust Combined Limits	65400	LM HIHI T-Exhaust Combined A	520525	16	3	1	68
T-Exhaust Combined Limits	65400	LM HI T-Exhaust Combined B	520526	16	5	1	68
T-Exhaust Combined Limits	65400	LM HIHI T-Exhaust Combined B	520527	16	7	1	68
T-Winding Limits	65419	LM HIHI T-Winding L1	520726	8	1	1	67
T-Winding Limits	65419	LM HI T-Winding L1	520727	8	2	1	67
T-Winding Limits	65419	LM HIHI T-Winding L3	520728	8	3	1	67
T-Winding Limits	65419	LM HI T-Winding L3	520729	8	4	1	67
T-Winding Limits	65419	LM HIHI T-Winding L2	520730	8	5	1	67
T-Winding Limits	65419	LM HI T-Winding L2	520731	8	6	1	67

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	Material-Nr./Material No.	XZ00E52000037	
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Aenderungsbeschreibung/Description of Revision war kZ				Kommt vor/Frequency																																																																																																																																																																																																																																																																																												
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<table><thead><tr><th>PGN-Name</th><th>PGN</th><th>SPN-Name</th><th>SPN</th><th>BitSize</th><th>StartByte</th><th>StartBit</th><th>SLOT</th></tr></thead><tbody><tr><td>T-Winding Limits</td><td>65419</td><td>LM LOLO P-Lube Oil (EMU)</td><td>520723</td><td>8</td><td>7</td><td>1</td><td>47</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM LO P-Lube Oil (EMU)</td><td>520724</td><td>8</td><td>8</td><td>1</td><td>47</td></tr><tr><td>MTU Generator Temperatures 1</td><td>65445</td><td>T-Coolant Gen Out</td><td>521810</td><td>8</td><td>3</td><td>1</td><td>68</td></tr><tr><td>MTU Generator Temperatures 1</td><td>65445</td><td>T-Air Inlet Gen</td><td>521839</td><td>8</td><td>1</td><td>1</td><td>67</td></tr><tr><td>MTU Generator Temperatures 1</td><td>65445</td><td>T-Air Outlet Gen</td><td>521840</td><td>8</td><td>2</td><td>1</td><td>67</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>Ready For Start Stop FMCS</td><td>520718</td><td>2</td><td>1</td><td>7</td><td>87</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>Engine Stop Active</td><td>520719</td><td>2</td><td>2</td><td>1</td><td>87</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>Start Active</td><td>520720</td><td>2</td><td>2</td><td>3</td><td>87</td></tr><tr><td>MTU Generator Temperatures 1</td><td>65445</td><td>T-Coolant Gen In</td><td>521811</td><td>8</td><td>4</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-Bearing GenDE</td><td>521816</td><td>8</td><td>21</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-Bearing GenDE</td><td>521817</td><td>8</td><td>22</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-Bearing GenNDE</td><td>521818</td><td>8</td><td>23</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-Bearing GenNDE</td><td>521819</td><td>8</td><td>24</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-AirInlet Gen</td><td>521820</td><td>8</td><td>17</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-AirInlet Gen</td><td>521821</td><td>8</td><td>18</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-AirOutlet Gen</td><td>521822</td><td>8</td><td>19</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-AirOutlet Gen</td><td>521823</td><td>8</td><td>20</td><td>1</td><td>68</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>FIFI Mode Active</td><td>521824</td><td>2</td><td>3</td><td>1</td><td>87</td></tr><tr><td>MTU Binary Signals</td><td>65420</td><td>Cmd Generator Shutdown</td><td>521825</td><td>2</td><td>3</td><td>3</td><td>87</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-Coolant Gen Out</td><td>521812</td><td>8</td><td>13</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI T-Coolant Gen In</td><td>521813</td><td>8</td><td>14</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-Coolant Gen Out</td><td>521814</td><td>8</td><td>15</td><td>1</td><td>68</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI T-Coolant Gen In</td><td>521815</td><td>8</td><td>16</td><td>1</td><td>68</td></tr><tr><td>Dash Display 2</td><td>64671</td><td>Engine Oil Filter Differential Pressure (Extended)</td><td>6321</td><td>16</td><td>1</td><td>1</td><td>52</td></tr><tr><td>MTU Eng Contr 4 - Addi Inform</td><td>65284</td><td>Engine Load Reserve</td><td>521826</td><td>8</td><td>8</td><td>1</td><td>44</td></tr><tr><td>Engine Electronic Controller #1</td><td>61444</td><td>Injection Quantity 0-120</td><td>512</td><td>8</td><td>2</td><td>1</td><td>44</td></tr><tr><td>MTU Eng Fluid Level Pressure 1</td><td>65363</td><td>P-Exhaust Back</td><td>521828</td><td>16</td><td>6</td><td>1</td><td>52</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HI P-Exhaust Back</td><td>521829</td><td>16</td><td>9</td><td>1</td><td>52</td></tr><tr><td>T-Winding Limits</td><td>65419</td><td>LM HIHI P-Exhaust Back</td><td>521830</td><td>16</td><td>11</td><td>1</td><td>52</td></tr><tr><td>MTU Several Limits 2</td><td>65448</td><td>LM AL RA Tank Empty</td><td>521933</td><td>8</td><td>1</td><td>1</td><td>44</td></tr><tr><td>Aftertreatment 2 SCR Exhaust Gas Pressures</td><td>64825</td><td>Aftertreatment 2 SCR Differential Pressure</td><td>4411</td><td>16</td><td>1</td><td>1</td><td>85</td></tr><tr><td>Aftertreatment 2 SCR Exhaust Gas Pressures</td><td>64825</td><td>Aftertreatment 2 SCR Intake Pressure</td><td>6587</td><td>16</td><td>4</td><td>1</td><td>355</td></tr><tr><td>MTU Energy Information</td><td>65428</td><td>Maximum Available Power</td><td>521057</td><td>16</td><td>5</td><td>1</td><td>55</td></tr><tr><td>Aftertreatment 1 Diesel Exhaust Fluid Supply Information</td><td>64828</td><td>Aftertreatment 1 Diesel Exhaust Fluid Pump</td><td>7107</td><td>2</td><td>6</td><td>5</td><td>87</td></tr></tbody></table>									PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT	T-Winding Limits	65419	LM LOLO P-Lube Oil (EMU)	520723	8	7	1	47	T-Winding Limits	65419	LM LO P-Lube Oil (EMU)	520724	8	8	1	47	MTU Generator Temperatures 1	65445	T-Coolant Gen Out	521810	8	3	1	68	MTU Generator Temperatures 1	65445	T-Air Inlet Gen	521839	8	1	1	67	MTU Generator Temperatures 1	65445	T-Air Outlet Gen	521840	8	2	1	67	MTU Binary Signals	65420	Ready For Start Stop FMCS	520718	2	1	7	87	MTU Binary Signals	65420	Engine Stop Active	520719	2	2	1	87	MTU Binary Signals	65420	Start Active	520720	2	2	3	87	MTU Generator Temperatures 1	65445	T-Coolant Gen In	521811	8	4	1	68	T-Winding Limits	65419	LM HI T-Bearing GenDE	521816	8	21	1	68	T-Winding Limits	65419	LM HIHI T-Bearing GenDE	521817	8	22	1	68	T-Winding Limits	65419	LM HI T-Bearing GenNDE	521818	8	23	1	68	T-Winding Limits	65419	LM HIHI T-Bearing GenNDE	521819	8	24	1	68	T-Winding Limits	65419	LM HI T-AirInlet Gen	521820	8	17	1	68	T-Winding Limits	65419	LM HIHI T-AirInlet Gen	521821	8	18	1	68	T-Winding Limits	65419	LM HI T-AirOutlet Gen	521822	8	19	1	68	T-Winding Limits	65419	LM HIHI T-AirOutlet Gen	521823	8	20	1	68	MTU Binary Signals	65420	FIFI Mode Active	521824	2	3	1	87	MTU Binary Signals	65420	Cmd Generator Shutdown	521825	2	3	3	87	T-Winding Limits	65419	LM HI T-Coolant Gen Out	521812	8	13	1	68	T-Winding Limits	65419	LM HI T-Coolant Gen In	521813	8	14	1	68	T-Winding Limits	65419	LM HIHI T-Coolant Gen Out	521814	8	15	1	68	T-Winding Limits	65419	LM HIHI T-Coolant Gen In	521815	8	16	1	68	Dash Display 2	64671	Engine Oil Filter Differential Pressure (Extended)	6321	16	1	1	52	MTU Eng Contr 4 - Addi Inform	65284	Engine Load Reserve	521826	8	8	1	44	Engine Electronic Controller #1	61444	Injection Quantity 0-120	512	8	2	1	44	MTU Eng Fluid Level Pressure 1	65363	P-Exhaust Back	521828	16	6	1	52	T-Winding Limits	65419	LM HI P-Exhaust Back	521829	16	9	1	52	T-Winding Limits	65419	LM HIHI P-Exhaust Back	521830	16	11	1	52	MTU Several Limits 2	65448	LM AL RA Tank Empty	521933	8	1	1	44	Aftertreatment 2 SCR Exhaust Gas Pressures	64825	Aftertreatment 2 SCR Differential Pressure	4411	16	1	1	85	Aftertreatment 2 SCR Exhaust Gas Pressures	64825	Aftertreatment 2 SCR Intake Pressure	6587	16	4	1	355	MTU Energy Information	65428	Maximum Available Power	521057	16	5	1	55	Aftertreatment 1 Diesel Exhaust Fluid Supply Information	64828	Aftertreatment 1 Diesel Exhaust Fluid Pump	7107	2	6	5	87
PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM LOLO P-Lube Oil (EMU)	520723	8	7	1	47																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM LO P-Lube Oil (EMU)	520724	8	8	1	47																																																																																																																																																																																																																																																																																									
MTU Generator Temperatures 1	65445	T-Coolant Gen Out	521810	8	3	1	68																																																																																																																																																																																																																																																																																									
MTU Generator Temperatures 1	65445	T-Air Inlet Gen	521839	8	1	1	67																																																																																																																																																																																																																																																																																									
MTU Generator Temperatures 1	65445	T-Air Outlet Gen	521840	8	2	1	67																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	Ready For Start Stop FMCS	520718	2	1	7	87																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	Engine Stop Active	520719	2	2	1	87																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	Start Active	520720	2	2	3	87																																																																																																																																																																																																																																																																																									
MTU Generator Temperatures 1	65445	T-Coolant Gen In	521811	8	4	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-Bearing GenDE	521816	8	21	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-Bearing GenDE	521817	8	22	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-Bearing GenNDE	521818	8	23	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-Bearing GenNDE	521819	8	24	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-AirInlet Gen	521820	8	17	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-AirInlet Gen	521821	8	18	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-AirOutlet Gen	521822	8	19	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-AirOutlet Gen	521823	8	20	1	68																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	FIFI Mode Active	521824	2	3	1	87																																																																																																																																																																																																																																																																																									
MTU Binary Signals	65420	Cmd Generator Shutdown	521825	2	3	3	87																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-Coolant Gen Out	521812	8	13	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI T-Coolant Gen In	521813	8	14	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-Coolant Gen Out	521814	8	15	1	68																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI T-Coolant Gen In	521815	8	16	1	68																																																																																																																																																																																																																																																																																									
Dash Display 2	64671	Engine Oil Filter Differential Pressure (Extended)	6321	16	1	1	52																																																																																																																																																																																																																																																																																									
MTU Eng Contr 4 - Addi Inform	65284	Engine Load Reserve	521826	8	8	1	44																																																																																																																																																																																																																																																																																									
Engine Electronic Controller #1	61444	Injection Quantity 0-120	512	8	2	1	44																																																																																																																																																																																																																																																																																									
MTU Eng Fluid Level Pressure 1	65363	P-Exhaust Back	521828	16	6	1	52																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HI P-Exhaust Back	521829	16	9	1	52																																																																																																																																																																																																																																																																																									
T-Winding Limits	65419	LM HIHI P-Exhaust Back	521830	16	11	1	52																																																																																																																																																																																																																																																																																									
MTU Several Limits 2	65448	LM AL RA Tank Empty	521933	8	1	1	44																																																																																																																																																																																																																																																																																									
Aftertreatment 2 SCR Exhaust Gas Pressures	64825	Aftertreatment 2 SCR Differential Pressure	4411	16	1	1	85																																																																																																																																																																																																																																																																																									
Aftertreatment 2 SCR Exhaust Gas Pressures	64825	Aftertreatment 2 SCR Intake Pressure	6587	16	4	1	355																																																																																																																																																																																																																																																																																									
MTU Energy Information	65428	Maximum Available Power	521057	16	5	1	55																																																																																																																																																																																																																																																																																									
Aftertreatment 1 Diesel Exhaust Fluid Supply Information	64828	Aftertreatment 1 Diesel Exhaust Fluid Pump	7107	2	6	5	87																																																																																																																																																																																																																																																																																									
 A Rolls-Royce solution Rolls-Royce Solutions GmbH		Zeichnungs-Nr./Drawing No. Material-Nr./Material No.		ZNG00045266 XZ00E52000037			Blatt/ Sheet 9 von/of 42																																																																																																																																																																																																																																																																																									
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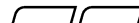
Aenderungsbeschreibung/Description of Revision war kZ			Kommt vor/Frequency
Buchst./Rev.Ltr. -4	Erh-Antrags-Nr. / Promotion request No. PR085807	Aenderungs-Nr./Revision Notice No.	Bearbeitungsstatus/Lifecycle Freigegeben

J1939 SPN Rx Signals

PGN-Name	PGN	SPN-Name	SPN	BitSize	StartByte	StartBit	SLOT
MTU System Operating PDU1 prop A1	61184	Engine Start Command	520192	2	1	1	87
MTU System Operating PDU1 prop A1	61184	Engine Stop Command	520193	2	1	3	87
MTU System Operating PDU1 prop A1	61184	Engine Safety and Protection Override Command	520194	2	1	5	87
MTU System Operating PDU1 prop A1	61184	MTU Stop Priming Command	520213	2	1	7	87
MTU System Operating PDU1 prop A1	61184	MTU DM1 Global DTC Acknowledge Command	520265	2	2	1	87
MTU System Operating PDU1 prop A1	61184	MTU DM1 Horn Command	520816	2	2	3	87
MTU System Operating PDU1 prop A1	61184	MTU RESET Trip Command	988	2	2	7	87
MTU System Operating PDU1 prop A1	61184	Engine Turn Command	520196	2	3	1	87
MTU System Operating PDU1 prop A1	61184	Engine Overspeed Test Command	520197	2	3	3	87
MTU System Operating PDU1 prop A1	61184	Harbour Test Mode Command	520199	2	3	5	87

J1939 Slot List NMEA2000 Data Format

SLOT Identifier	SLOT Name	SLOT Type	Scaling	Range	Offset	Length
1	SAEpr11	Pressure	5 kPa/bit	0 to 1,250 kPa	0	1 byte
2	SAEpr13	Pressure	8 kPa/bit	0 to 2,000 kPa	0	1 byte
3	SAEtm11	Time	1 hr/bit	0 to 250 hr	0	1 byte
4	SAEtm10	Time	1 hr/bit	-125 to 125 hr	-125 hr	1 byte
5	SAEtm12	Time	1 hr/bit	-32,127 to 32,128 hr	-32,127 hr	2 bytes
6	SAEtm06	Time	1 s/bit	0 to 4,211,081,215 s	0	4 bytes
7	SAEad01	Angle/Direction	10^-7 deg/bit	-210 to 211.1081215 deg	-210 deg	4 bytes
8	SAEad04	Angle/Direction	1/128 deg/bit	-200 to 301.99 deg	-200 deg	2 bytes
9	SAEad05	Angle/Direction	1/128 deg/bit	0 to 501.99 deg	0	2 bytes
10	SAEad02	Angle/Direction	0.1 sec/bit	-3276.8 to 3148.7 sec	-3,276.8 sec	2 bytes
11	SAEad06	Angle/Direction	1 deg/bit	-125 to 125 deg	-125 deg	1 byte
12	SAEct04	Count	1 count/bit	0 to 64,255	0	2 bytes
13	SAEds03	Distance	0.1 mm/bit	-3,200 to 3,225.5 mm	-3,200 mm	2 bytes
14	SAEds04	Distance	0.1 mm/bit	0 to 6,425.5 mm (0 to 6.4255 m)	0	2 bytes
15	SAEds08	Distance	1 m/bit	-125 to 125 m	-125 m	1 byte
16	SAEel01	Economy, liquid	1/512 km/L per bit	0 to 125.498046875 km/L	0	2 bytes
17	SAEeg01	Economy, gaseous	1/512 km/kg per bit	0 to 125.498046875 km/kg	0	2 bytes
18	SAEfg01	Flow rate, gaseous	0.05 kg/h per bit	0 to 3212.75 kg/h	0	2 bytes
19	SAEfv01	Flow rate, volumetric	0.1 m^3/h per bit	0 to 6425.5 m^3/h	0	2 bytes
20	SAEfu01	Fuel Used, liquid	0.5 L/bit	0 to 2,105,540,607.5 L	0	4 bytes
21	SAEfu01	Fuel Used, gaseous	0.5 kg/bit	0 to 2,105,540,607.5 kg	0	4 bytes
22	SAEgg01	Governor gain	1/1280 %/rpm per bit	0 to 50.2 %/rpm	0	2 bytes
23	SAEgr01	Gear Ratio	0.01/bit	0 to 642.55	0	2 bytes
24	SAEgv01	Gear Value	1 gear value/bit	-125 to 125	-125	1 byte
25	SAEid01	Identifier, component/software	1 ID/bit	0 to 250 ID	0	1 byte
26	SAEkv01	Kinematic viscosity	1 mm^2/s per bit	0 to 250 mm^2/s	0	1 byte
27	SAEma11	Mass	0.5 kg/bit	0 to 32,127.5 kg	0	2 bytes
28	SAEma02	Mass	2 kg/bit	0 to 128,510 kg	0	2 bytes
29	SAEma03	Mass	10 kg/bit	0 to 642,550 kg	0	2 bytes
30	SAEpc01	Percent, position/level	0.0025 %/bit	0 to 160.6375 %	0	2 bytes
31	SAEr02	Ratio	0.1/bit	0 to 25.0	0	1 byte
32	SAEr01	Ratio	0.001/bit	0 to 64.255	0	2 bytes
33	SAEre01	Record	1 record/bit	0 to 250 records	0	1 byte
34	SAErV01	Revolutions	1000 r/bit	0 to 4,211,081,215,000 r	0	4 bytes
35	SAEsa01	Source Address	1 source address/bit	0 to 255	0	1 byte
36	SAEds10	Distance	0.125 km/bit	0 to 526,385,151.875 km	0	4 bytes
37	SAEds11	Distance	5 km/bit	-160,635 to 160,640 km	-160,635 km	2 bytes
38	SAEds09	Distance	5 m/bit	0 to 21,055,406 km	0	4 bytes
39	SAEds06	Distance	0.125 m/bit	-2500 to 5531.875 m	-2500 m	2 bytes
40	SAEec02	Electrical Current	1 A/bit	-125 to 125 A	-125 A	1 byte
41	SAEec03	Electrical Current	1 A/bit	0 to 250 A	0	1 byte
42	SAEfl01	Flow rate, liquid	0.05 L/h per bit	0 to 3,212.75 L/h	0	2 bytes
43	SAEfr01	Force	5 N/bit	0 to 321,275 N	0	2 bytes
44	SAEpc03	Percent, position/level	0.4 %/bit	0 to 100 %	0	1 byte
45	SAEpc05	Percent, position/level	1 %/bit	-125 to 125 %	-125 %	1 byte
46	SAEpc06	Percent, position/level	1 %/bit	0 to 250 %	0	1 byte

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	Material-Nr./Material No.	XZ00E52000037	
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